FinAI

Advanced AI Based Personal Finance & Investment Analysis Tool

Revision 3 · Market-adjusted Stock Fair Value

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Where the project stands

Four-phase roadmap — R3 completes the investment-analytics core

Phase 1	Foundation & core finance	**Done (R1)**
Phase 2a	Indicators · valuation · stock score	**Done (R2)**
Phase 3 (pulled fwd)	SIP / FD / XIRR calculators	**Done (R2)**
Phase 2b	News sentiment + market-adjusted Fair Value	**Done (R3)**
Phase 3	Portfolio risk · forecasting	**Next**
Phase 4	RAG assistant · hardening · deploy	**Planned**

The R3 question

"What is this stock actually worth right now?"

Fundamentals alone

DCF / P/E give a long-run value but ignore current trend, momentum and what the market is talking about.

Charts alone

Technicals capture momentum but have no notion of intrinsic worth — a stock can trend far above value.

News alone

Headlines move prices short term but are noisy and easy to over-react to.

R3 combines all three — with fundamentals as the anchor and every adjustment visible.

Market-adjusted Fair Value

backend/app/market/fair_value.py

Fair Value = Intrinsic Anchor × (1 + Technical tilt) × (1 + News-buzz tilt)

Intrinsic anchor

Mean of DCF (2-stage + Gordon TV), P/E, EV/EBITDA and DDM fair values from the R2 valuation engine.

Technical tilt · ±8%

Confluence of trend (price vs SMA50, SMA20/50 cross), RSI (overbought-aware), MACD histogram, Bollinger position.

News-buzz tilt · ±6%

Recency-weighted headline sentiment × buzz multiplier (high 1.0 · elevated 0.8 · normal 0.6).

Corridor = ±0.5 × annualised volatility (5–25%). Verdict: Undervalued · Fairly valued · Overvalued.

News buzz & sentiment engine

backend/app/market/sentiment.py

How a headline is scored

- Finance lexicon: words + phrases ("raises guidance", "margin expansion")
- Negation flips polarity ("not weak")
- Normalised: score = $x / \sqrt{(x^2 + 4)} \rightarrow (-1, 1)$
- Matched terms returned as evidence
- Recency decay, 7-day half-life
- Buzz level from 7-day article count

Honest limitations

- News is illustrative sample data, clearly labelled — not real headlines
- Lexicons miss context: "attrition declines" is good news but scores negative
- Next: Experiment 2 — lexicon vs TF-IDF+SVM vs FinBERT on a labelled set (accuracy / F1)
- Live news API plugs in behind load_news()

Why bounded tilts?

Design decision you can defend

Anchor stays fundamental

Technical + news can move value by at most ~14% combined. Short-term noise cannot dominate the estimate.

Fully explainable

API returns anchor, each tilt %, every technical signal and every scored headline. UI shows the waterfall.

Graceful degradation

No news → sentiment tilt = 0 and marked not_evaluated. Missing indicators are skipped, never invented.

What the user sees

Market page · new Fair Value panel (frontend/src/components/FairValuePanel.tsx)

Fair Value card

- Big fair-value figure + corridor
- Verdict badge (Undervalued / Fairly valued / Overvalued)
- Gauge: price marker vs fair-value marker inside corridor
- Waterfall: anchor → technical tilt → news tilt
- Signal chips: trend · RSI · MACD · Bollinger with values

News Buzz feed

- Overall sentiment + label
- 👍 / 👎 / neutral counts
- Each headline with score and matched terms

API, tests & running it

New endpoint

GET /api/v1/market/{symbol}/fair-value

/analysis now also returns fair_value + news

Tests

8 new tests in test_fair_value.py: polarity, negation, bounds, recency, caps, verdicts, endpoint, 404.

One-click

START-FINAI-R3.cmd → starts Docker, stops R2 if running, builds, seeds demo data, opens http://localhost:8080/app

STOP-FINAI-R3.cmd → stops containers, keeps data

Next steps

Toward Review 3 / final

Phase 2 close-out

- Live market + news provider
- Experiment 2: sentiment model comparison
- PDF research report

Phase 3

- Portfolio analyzer & risk (HHI, beta)
- Expense forecasting experiment (MA/Holt/ARIMA/GBM)
- Back-test the Fair Value verdicts

Phase 4

- RAG research assistant with citations
- Hardening, e2e tests, deployment
- Dissertation freeze

Thank you

Questions & live demo

Educational/analytical tool — not investment advice. Demo uses recorded sample data.